

HAT-P-12b

R-1298

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_210310 · created 2026-07-31T16:57:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459283.7598 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.068 +/- 0.054
Transit depth (Rp/Rs)^2	0.46 +/- 0.73 [%]
Orbital Inclination (inc)	86.67 +/- 1.71
Airmass coefficient 1 (a1)	0.7 +/- 0.36
Airmass coefficient 2 (a2)	0.25 +/- 0.17
Scatter in the residuals of the lightcurve fit is	53.29 %
Best Comparison Star	None
Optimal Aperture	2.41
Optimal Annulus	5.25
Transit Duration (day)	0.061 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.068 ± 0.054 vs 0.1406 (deviation 1.34σ)
Duration fit vs published	0.061 d vs 0.0975 d (deviation 1.18σ)
Mid-time O–C	-8.7 min
Depth SNR	1.3
Residual scatter	53.29 %

Light curve

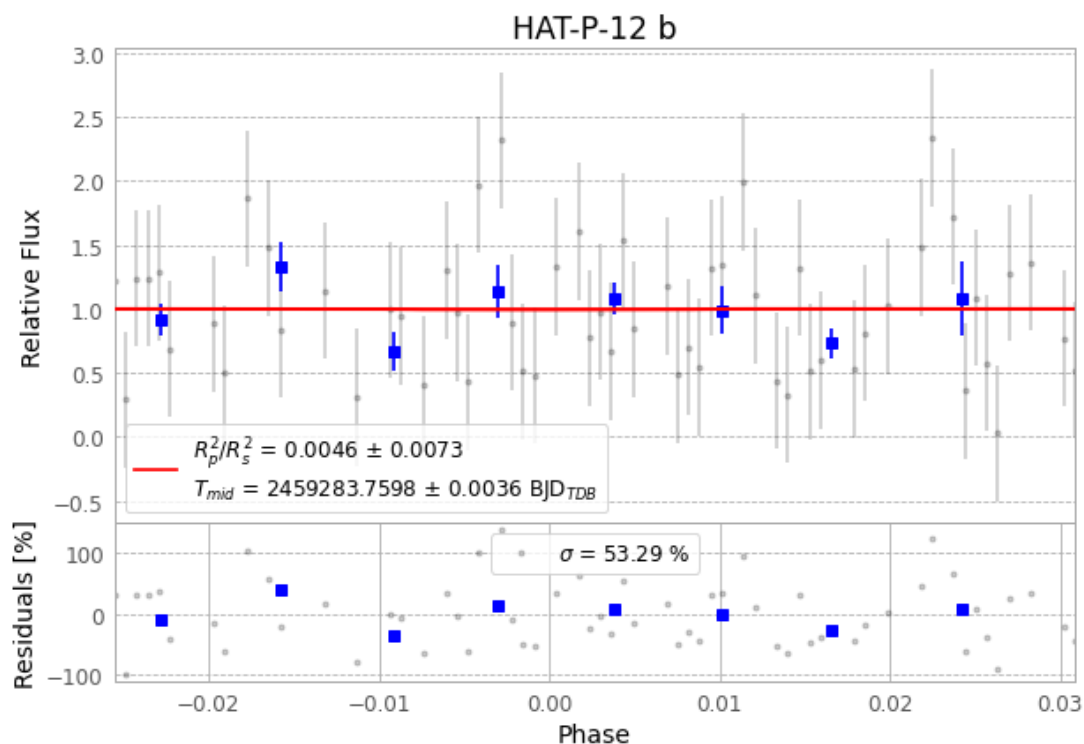


Figure 1 — FinalLightCurve_HAT-P-12 b_2021-03-09.png (SHA-256 9cc96909fbde1ba5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 125], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 86062e674c395c10...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

88 FITS frames (58 MB), HATP-12210310040912.FITS → HATP-12210310083012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-210310.log (9cebb21b9dab...), FinalLightCurve_HAT-P-12 b_2021-03-09.png (9cc96909fbde...), FinalLightCurve_HAT-P-12 b_2021-03-09.csv (46d010a6fd24...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 16:57Z.